Deep Learning

Project 3: Comparison of Deep Learning Architectures

Name: Soham Sen
Class: BCA Sem 5
Reg No: 2411021240008
Date: 30/08/2026

Introduction:

Deep Learning is a branch of Machine Learning that uses artificial neural networks with multiple layers to learn patterns from data. Different deep learning architectures are designed for different types of data and problems. This project focuses on three important architectures: Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs).

Basic Architecture:

1. Feedforward Neural Network (FNN)

A Feedforward Neural Network is one of the simplest types of neural networks. Information flows in only one direction, from the input layer through one or more hidden layers to the output layer.

Basic structure:

Input Layer → Hidden Layer(s) → Output Layer

Each neuron is connected to neurons in the next layer using weights and biases. Activation functions such as ReLU or Sigmoid help the network learn complex patterns.

2. Convolutional Neural Network (CNN)

A Convolutional Neural Network is mainly designed to process images and other spatial data. CNNs use convolutional layers to detect features such as edges, shapes, textures, and objects.

A typical CNN consists of:

Input → Convolution → Activation → Pooling → Fully Connected Layer → Output

- Convolution: Detects important features from the input.
- Activation: Introduces non-linearity, commonly using ReLU.
- Pooling: Reduces the spatial size of the feature maps.
- Fully Connected Layer: Uses the extracted features to make a prediction.

3. Recurrent Neural Network (RNN)

A Recurrent Neural Network is designed for sequential or time-dependent data. Unlike a feedforward network, an RNN can use information from previous inputs through its hidden state.

Basic structure:

Input → Hidden State → Output

The hidden state carries information from previous time steps, allowing the network to understand sequences.

RNNs are commonly used for text, speech, and time-series data. More advanced versions such as LSTM and GRU are used to handle long-term dependencies more effectively.

Suitable Applications:

Feedforward Neural Networks

Feedforward networks are suitable for:

- Classification of structured/tabular data
- Regression problems
- House price prediction
- Customer classification
- Credit-risk prediction
- Basic pattern recognition

Example: Predicting whether a customer will leave a company based on their age, income, and usage data.

Convolutional Neural Networks

CNNs are suitable for:

- Image classification
- Face recognition
- Object detection
- Medical image analysis
- Image segmentation
- Computer vision

Example: A CNN can analyze an X-ray image and classify it as normal or abnormal.

Recurrent Neural Networks

RNNs are suitable for:

- Natural Language Processing (NLP)
- Speech recognition
- Text prediction
- Time-series forecasting

- Sequence prediction
- Sentiment analysis

Example: An RNN can analyze a sequence of words to predict the sentiment of a sentence.

Comparison of Architectures

Feature	Feedforward Neural Network	CNN	RNN
Structure	Input → Hidden → Output	Convolution → Pooling → Fully Connected	Recurrent layers with hidden state
Main Input	Fixed-size/tabular data	Images and spatial data	Sequential/time-dependent data
Memory	No memory of previous inputs	No inherent temporal memory	Uses previous information through hidden state
Main Strength	Simple prediction and classification	Learning spatial features	Learning sequential patterns
Important Operation	Weighted connections	Convolution/filtering	Recurrent connections
Common Applications	Classification, regression	Image recognition, object detection	Text, speech, time series
Example	House price prediction	Identifying objects in an image	Predicting the next word in a sentence

Key Differences

Feedforward Neural Networks:
Best for fixed and structured data where the order of the input is not important.

CNNs:
Best for images and spatial data because they can automatically learn visual features such as edges, shapes, and objects.

RNNs:
Best for sequential data where the order of the information matters, such as text, speech, and time-series data.

Conclusion

Feedforward Neural Networks, CNNs, and RNNs are important deep learning architectures, but each is designed for different types of problems. FNNs are commonly used for structured data and general prediction tasks. CNNs are highly effective for images and computer vision because they learn spatial features. RNNs are suitable for sequential data because they can use information from previous time steps.

Therefore, the choice of architecture depends mainly on the type of input data and the problem being solved. Understanding these differences helps us select the most appropriate neural network for a particular real-world application.